Budget Allocation — Quick Reference Notes

1. Compression ratio — factorised vs. dense

- **Dense parameters**: the original weight matrix has $d_h \times d_i$ parameters.
- **Factorised parameters**: a rank-k SVD splits it into two matrices:

$$\underbrace{(d_h \times k)}_{\text{left factor}} + \underbrace{(k \times d_i)}_{\text{right factor}} = k\,(d_h + d_i).$$

- **Compression ratio**:

$$\alpha = \frac{\text{factorised params}}{\text{dense params}} = \frac{k\,(d_h + d_i)}{d_h \cdot d_i}.$$

  So $\alpha = 0.80$ means "keep 80% of the original parameter count."

2. Why that formula for k

Just solve $\alpha = \dfrac{k\,(d_h + d_i)}{d_h \cdot d_i}$ for k:

$$\boxed{k_{\text{uniform}} = \alpha \cdot \frac{d_h \cdot d_i}{d_h + d_i}}$$

Nothing deep — it is the rank that achieves exactly the target parameter ratio.

3. Redistribution formula

Given L layers with the same total budget $k_{\text{uniform}} \times L$, each layer ℓ receives rank proportional to its protection score:

$$\boxed{k_\ell = \big(k_{\text{uniform}} \times L\big) \cdot \frac{\text{protection\_score}_\ell}{\sum_{j=1}^{L} \text{protection\_score}_j}}$$

Total budget is identical to the uniform case; layers with higher protection scores get proportionally more rank.

4. Why the upper bound $k \le d_h$

The gate_proj and up_proj matrices have shape $d_i \times d_h$. A rank-k approximation cannot exceed the smaller dimension, which is d_h. So $k > d_h$ is mathematically impossible — you would be requesting more singular values than exist. The clamp prevents the proportional formula from assigning nonsensical ranks to high-protection layers.

5. How rounding adjustment works

The proportional formula yields float ranks (e.g. 614.3). The adjustment procedure:

1. Floor every k_ℓ to an integer. The integer total will be short of the budget by some amount δ.
2. Sort layers by **fractional part** (descending): a layer with fractional part 0.9 gets +1 before a layer with 0.1.
3. Break ties by protection score, then by raw rank.
4. Add +1 to the top-δ layers until the integer sum equals $k_{\text{uniform}} \times L$ exactly.

This is essentially banker's rounding: give extra rank units to layers that were closest to rounding up anyway, with protection score as tiebreaker.